UTKARSH®
Life demands. We deliver.

Arrjavv Square, 95A, Elliot Road, 4th Floor,
Kolkata - 700 016, India

www.utkarshindia.in

For Enquiry call: 033-4190-0001

UTKARSH®
Life demands. We deliver.



Noise control, engineered with precision.

Where innovation meets application in noise control.

NOISE BARRIER

Noise is characterised by irregular vibrations that are considered unpleasant and cause disturbance. To reduce noise levels that disrupt daily life, Utkarsh India has engineered Noise Barriers, leveraging modern technology to deliver world-class solutions.

There are two types of noise barriers that help lower decibel levels— Reflective Barriers, made from transparent or translucent polycarbonate sheets and Absorptive Barriers, constructed with rockwool or glasswool sandwiched between metal sheets. The application areas and costs may vary depending on the material type and specific project requirements.



Types of Noise Barriers

Reflective Noise Barrier

Types

1. Solid Polycarbonate sheet
2. Multiwall Polycarbonate sheet



Features

- Thickness varies from **8 mm - 20 mm**
- Sound transmission class (STC) – **18dB - 26 dB**
- Airborne Sound Insulation (DLR) – **18 dB - 30 dB**

Cost

- Depends majorly on the polycarbonate/acrylic sheet
- Hollow sheets (MW sheet) are on the lowest side
- Solid sheet's rate increases with its thickness

Absorptive Noise Barrier

Types

1. Mono-absorbent
2. Bi-absorbent



Features

- Thickness varies from **100 mm - 150 mm**
- Sound transmission class (STC) – **20dB - 45 dB**
- Airborne Sound Insulation (DLR) – **18 dB - 45 dB**

Cost

- Here the quality & dimension of the rockwool plays an important role, along with the structural design of the barrier

It's time to cancel the noise and ensure peace of mind, powered by precision and innovation

Main Purposes of Transparent Noise Barriers

- Create noise barriers between high traffic and populated areas
- Maintain visibility and landscape sequence
- Provide lightweight and efficient solution for bridges, in comparison with complete concrete designs
- Provide lightweight extensions for concrete walls

Advantages of Transparent Noise Barrier

- Low weight: Less than half the weight of glass with similar thickness
- High transparency: Matching that of the clearest glass
- Durability: High resistance to impact and vandalism (PALSUN)
- Weather ability: Excellent resistance to UV radiation and extreme outdoor conditions
- Flexibility that allows curving, bending and shaping without any special treatment
- Compliance with international standards
- Fire resistance
- Acoustic comfort
- Sustainable materials



Major factors should be considered while designing noise barriers

- **Sound barrier location**
- **Material**
- **Design and shape of barrier**
- **Wind speed calculations**
- **Noise calculations**
- **Structural stability**
- **Aesthetic values**

Specifications (Absorptive Noise Barrier)

Charecteristic	Specification
Panel sizes	3960 x 500 mm, 2960 x 500 mm, 2460 x 500 mm, 1960 x 500 mm
Panel thickness	80 mm | 100 mm | 120 mm
Wind load	Designed as per is 875 part-iii, 1987
Sound proofing	Class A4
Sound deadening	Class B2
Type of wall	Double skin insulated wall
Facing sheet	Plain metal sheet/GI sheet/aluminium, 0.5 to 1.2 mm thick
Backside sheet	Perforated sheet/GI sheet/aluminium with 22-46 % open area
Infill/core	Rockwool confirming to IS-8183 & BS 476 of varies density and Thickness. Rockwool covered with fiber tissue paper fire- resistive.
Cover	Water-repellent tissue paper for infill material lamination, 50-micron thickness
Sound damping	High density mass loaded vinyl barrier of 2 mm thickness
Joint	Each panel having a tongue & groove to join perfectly witheach other
Gasket	The panel and structure having neoprene rubber gaskets at the Joint to block noise leakage
Vertical post	The structure is of 'H' profile in which acoustic structure panel slides Longitudinally
Finish	All metallic surface/parts are galvanised/ powder coated/painted in desire shade

Typical Sound Level Calculations

It is important to keep in mind that you should really be concerned with the noise level at the ear, instead of the noise that is actually produce by the equipment.

Sound or noise example	Sound level	Comment
Human ear	**0 dB**	This is a matter of definition
Rustle of leaf	**20 dB**	
Whisper	**30 dB**	
Normal conversation	**60 dB**	
Inside passenger car at 60 MPH	**65-75 dB**	Smaller cars are usually noisier
Ringing telephone	**80 dB**	Measured close to the phone
Hair dryer	**85-90 dB**	Measured close to the dryer
Lawn mower	**85-90 dB**	Measured at the ear
Maximum exposure, 8 hours	**90 dB**	Osha regulation
Chain saw	**90-100 dB**	Measured at ear, sound level varies by model
Tractor/Bulldozer	**95-105 dB**	Inside cab may be less noisy (Sound level varies by model)
Rock drill	**110-115 dB**	
Circular saw, table saw	**110 dB**	
Threshold of pain	**120 dB**	A matter of definition
Pneumatic chipper	**120 dB**	
Fire engine siren	**120 dB**	
ET engine at takeoff	**140 dB**	
12-gauge shotgun	**165 dB**	Impact sound uses a different risk criteria
Rocket launch	**180 dB**	
Loudest possible sound	**195 dB**	

